

Summary

This study examines historical Olympic data to predict medal outcomes for the 2028 Los Angeles Olympics. At the core of the analysis is a novel competition model, inspired by the Lotka-Volterra species competition and Bass market competition models. This innovative approach considers a wide range of factors influencing team performance, including innovation, competitiveness, and imitation. To enhance this model, a custom scoring algorithm is developed, which evaluates teams based on historical performance, event-specific dominance, and athlete experience. This comprehensive framework allows for more nuanced predictions of medal counts and highlights actionable strategies, such as the potential impact of investing in ‘great coaches’. We suggest ways three specific countries could significantly benefit from such investments ahead of the 2028 Olympics.

The competition model’s results demonstrate clear improvements over a simpler linear regression model, which was also tested in this study. While the linear regression model and other regression models provide reasonable predictions for total and gold medal counts, they fail to capture certain subtleties of Olympic performance dynamics, such as interactions between teams and the fixed nature of total medals available. By addressing these considerations, the competition model delivers more accurate and insightful forecasts. Furthermore, additional methods, including Random Forest and difference equations, are used to account for first-time medal winners and the broader effects of coaching strategies, improving the overall analysis.

Viewing the Olympics as a competition for fixed resources and incorporating sensitivity analyses increases the robustness of the results. This innovative approach provides a reliable framework for understanding and forecasting Olympic success.

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1 Introduction

The Olympic Games unite nations across the globe in the pursuit of excellence, camaraderie, and national pride. Every four years, the Summer Olympics present an opportunity to showcase athletic prowess, and the highly anticipated "medal table" serves as a measure of success for participating countries. Beyond those great powers, the games also offer a platform for smaller countries to achieve their first breakthroughs, highlighting the evolving dynamics of international competition. In this study, we aim to address several critical questions surrounding Olympic medal forecasting:

- Medal Predictions for 2028: Which countries are expected to improve or decline in their performance compared to Paris 2024? How can we quantify uncertainty in these predictions?
- First-Time Medalists: What are the chances that countries with no prior Olympic medals will achieve their first success, and which factors contribute most to this likelihood?
- Event and Medal Correlations: How do the number and types of events influence national medal counts? Which sports should nations prioritize to maximize their chances of success?
- The Impact of Coaching: To what extent does the ‘great coach effect’ influence medal outcomes? How can countries strategically leverage elite coaching in specific sports?

By developing robust mathematical models and leveraging historical Olympic data, this paper provides a data-driven framework to address these questions. Through careful analysis, we explore how nations can adapt their strategies to improve their medal standings and offer actionable insights to Olympic committees worldwide.

2 Analysis of the Problem

When predicting future data, traditional models such as time-series analysis or linear regression often rely solely on historical data to predict future medal counts for each country. While these approaches can capture trends and patterns over time, they fail to account for other subtleties or inherent constraints, like that the total number of medals is fixed in any given Olympic Games. Specifically, these models treat medal counts for each country independently, ignoring the fact that one country’s increase in medals directly reduces the amount available for others. This independence assumption leads to a fundamental inconsistency: the predicted total medal count across all nations often exceeds or falls short of the actual medal pool, violating the fixed-sum constraint of Olympic medals.

Furthermore, linear regression assumes a continuous, unbounded growth or decline in medal counts, which may not reflect the resource-limited and competitive nature of Olympic events. These limitations make traditional models inadequate for analyzing the zero-sum dynamics of medal competition, where countries’ performances are inherently interdependent. While it is useful to find a solution based on a naive approach like a regression model, a more suitable approach must explicitly consider these competitive interactions and the fixed medal pool, such as using a constrained or competitive modeling framework like the Lotka-Volterra and Bass models.

2.1 Data Processing

Given the historical complexities of Olympic participation, particularly in relation to geopolitical changes and evolving country definitions, it is necessary to adopt a systematic approach to prepare the data for analysis. The Olympic Games have a long history dating back to 1896, during which geopolitical shifts, wars, and changes in country designations have significantly impacted participation and medal counts. To mitigate the challenges posed by such changes, the following steps were taken: Data prior to 1914 was excluded due to inconsistencies in country participation, as the modern political landscape and country structures were not yet fully formed. Accordingly, *we only consider the Olympic Games from 1920 onward*. Additionally, early Olympic Games often had limited global representation, making comparisons with contemporary data less meaningful. Moreover, over the years, certain countries have undergone name changes, splits, or unifications (e.g., the dissolution of the Soviet Union and the formation of modern Germany). For consistency, countries were mapped to their modern designations where possible, ensuring alignment with the current geopolitical boundaries of the 2024 Paris Olympics.

The provided data files have many inconsistencies. For example, there are 22 teams that use different NOC codes in the `medal_counts` file and the `athletes` file, like how Burkina Faso uses the codes BUR and BFA. Other issues include some country names having a trailing space while others do not and some entries missing from the files. We search extensively through all the provided data files to ensure our results have the fewest number of avoidable errors possible.

2.2 Assumptions

- **Assumption 1:** Every team that participated in the Paris 2024 Olympics will compete in the Los Angeles 2028 Olympics. This is a fundamental assumption that provides a continuum for the data, allowing a country's past results to be an acceptable indicator for the future. There is one exception: we will assume Russian athletes will compete in the 2028 Olympics for the Russian national team (RUS). While this may not be very likely, it simplifies our preparations and results.
- **Assumption 2:** The medals earned by the ROC (Russian Olympic Committee) team in the Tokyo 2020/2021 Olympics count for Russia. This assumption allows us to consolidate the performance of Russian athletes under one team.

2.3 Initial Observations

Many interesting trends and relationships appear in the data provided. For instance, Figure 1 shows the mostly increasing trend in the number of countries that participate in the Olympics. Since 2000, the number of competing teams has been roughly constant. Figure 2 shows a similar trend but with the increasing behavior continuing while the number of competing countries slows down. This is to be expected with the increasing number of events and attention on the Olympics over the past century.

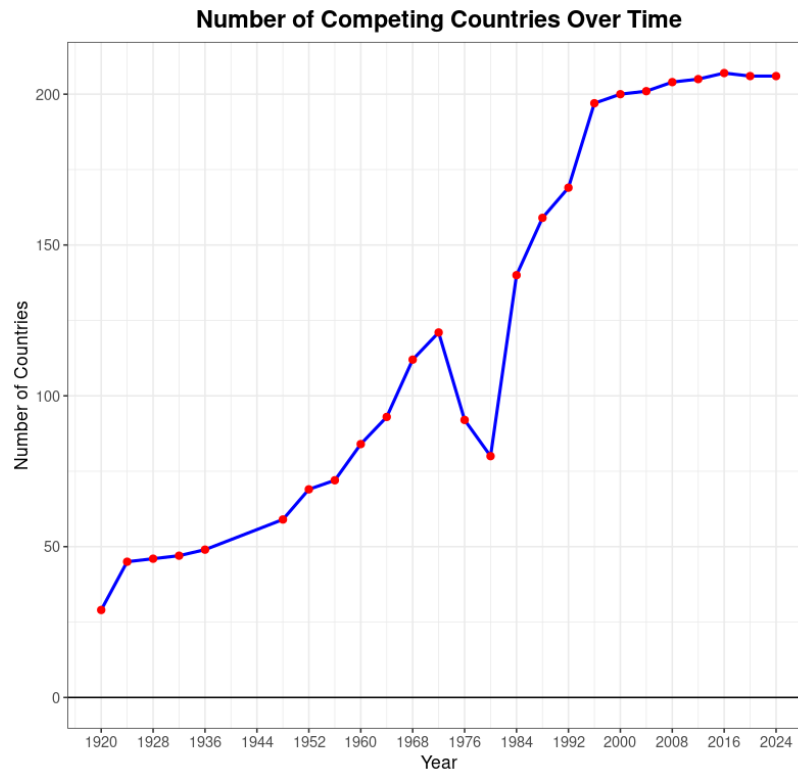


Figure 1: Number of Countries Competing Over Time

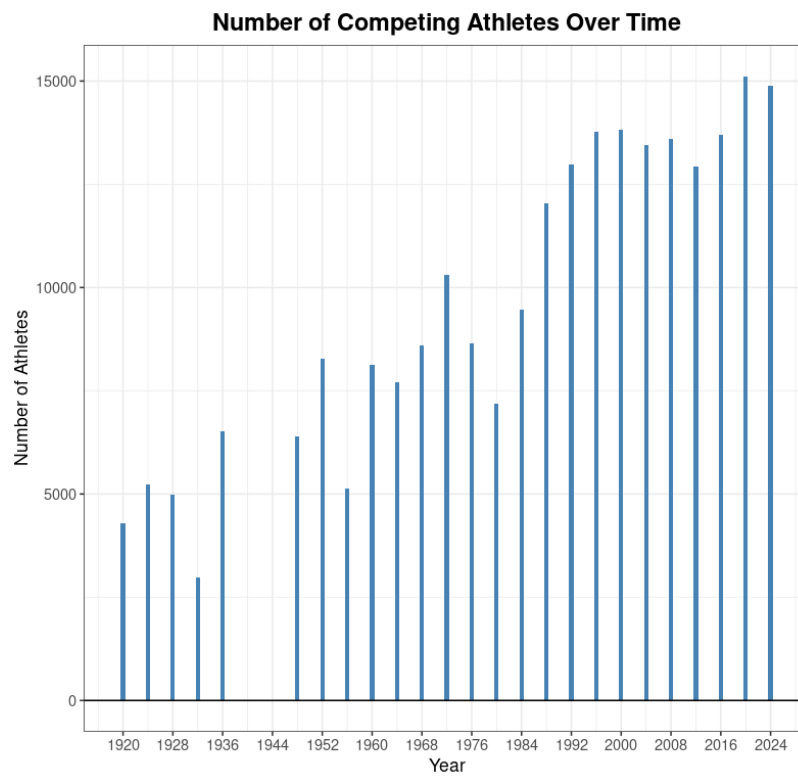


Figure 2: Number of Athletes Competing Over Time

2.4 Notation

Term	Definition
NOC	National Olympic Committee - Mostly equivalent to country codes, with a few exceptions
Team	Athletes competing under a single NOC
Non-individual event	An event where at least three athletes from the same NOC compete together

Table 1: Notation Table

3 Medal Predictions Models

3.1 Linear Regression Model

As a first attempt at predicting the 2028 medals distribution, we use linear regression models on the total number of medals and number of gold medals. We look for a linear relationship, where the number of medals is a function of the year and country. We standardized the Year variable to represent the number Olympics event, where the 1920 Olympics is Year 0 and the 2024 Olympics is Year 2026. As for the country, the linear model is automatically vectorized, giving small coefficients as contributions for each of the countries. This model does not address the question of first-time medalists, as it only considers countries that have won at least one medal in the past.

Our program solves for the coefficients in the equation

$$\text{Total Medals} = \beta_0 + \beta_1 \cdot \text{Year} + \beta_2 \cdot \text{Team}, \quad (1)$$

then computes 90% confidence intervals with R's `predict()` function.

As discussed earlier, a linear regression model is not the best choice for this problem. Indeed, a slightly upward trend may appear in the medal counts for countries over time, due in part to an increasing number of events, but the linear model is inherently unable to capture any other trends in the data. That said, this model's predictions are reasonable, correctly predicting nine of the top 10 when tasked with predicting results for 2024, and still useful for future comparison.

3.2 Adapted Lotka-Volterra-Bass Competition Model

3.2.1 Introduction to the ALVBC Model

The competitive nature and outcomes of the Olympic Games may, given the correct context, resemble ecological interactions. We consider the following situation:

- Teams are predators, each of a different species, competing with each other for a share of available resources, or prey
- Resources are the medals, where the total number of medals is constant throughout an Olympics

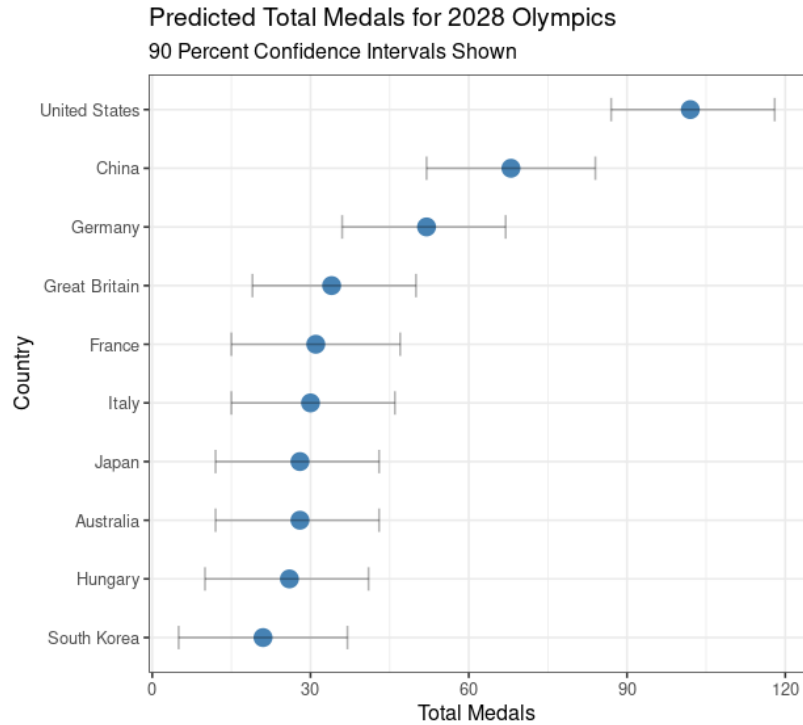


Figure 3: Linear Regression Prediction for Countries with the Top 10 Total Medals

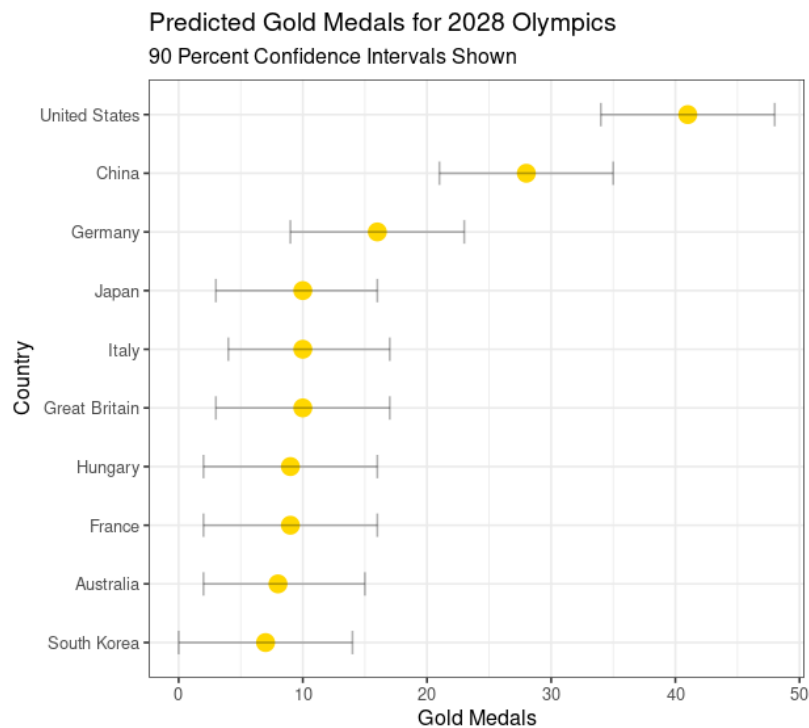


Figure 4: Linear Regression Prediction for Countries with the Top 10 Gold Medals

- Teams compete against each other differently, just as different animals hold unique advantages against others
- Teams are able to learn from past events and innovate their strategies, just as animals evolve and retain their strongest traits

Our model incorporates aspects of the competitive Lotka-Volterra model and the Bass diffusion model in order to accurately describe the Olympic Games. The competitive Lotka-Volterra equations[1] for i species form a system of i first-order differential equations describing logistic growth, defined as

$$\frac{dR_i}{dt} = r_i R_i \left(1 - \frac{R_i}{K_i}\right) - \sum_{j \neq i} \alpha_{ij} R_i R_j. \quad (2)$$

These equations describe the population dynamics of different species competing for a resource, making the crucial assumption that the *population of any species will increase in the absence of competition unless the population is already at the carrying capacity*[2]. Table 3.2.1 defines the terms in Equation 2 in an ecological context.

Term	Definition
R_i	The population of species i (as a function of time t)
r_i	The intrinsic growth rate of species i
K_i	The carrying capacity of species i
α_{ij}	The competition coefficient of species i on species j
$\sum_{j \neq i} \alpha_{ij} R_i R_j$	The total competitive effect of all other species j on species i

Table 2: Terms in the Competitive Lotka-Volterra equation

The Bass diffusion model[3] is an economic model that describes how new goods may spread within a market. The relationship is defined as

$$\frac{dN(t)_i}{dt} = p_i(m_i - N(t)_i) + q_i \frac{N(t)_i}{m_i} (m_i - N(t)_i). \quad (3)$$

Term	Definition
$N(t)_i$	The number of adopters of good i at time t
p_i	The coefficient of innovation of good i
q_i	The coefficient of imitation of good i
m_i	The total market potential of good i

Table 3: Terms in the Bass Diffusion Model

Connecting these two seemingly unrelated models is a matter of re-contextualizing the problem, this time in terms of goods in a market. Now, we consider the situation

- Team i winning a medal is equivalent to a consumer purchasing good i
- The total number of medals a team can possibly win is the total market potential of a good

- Coaches may imitate the strategies of successful teams. The degree of imitation is measured by q
- Teams learn from their experience, represented by the innovation coefficient p

Valla[1] combines the Lotka-Volterra and Bass models to represent pharmacological market competition. We further alter the model to represent team competition in the Olympic Games. The Lotka-Volterra-Bass Competition (LVBC) model is defined, using the same coefficients as Equation 2, as

$$\frac{dR_i}{dt} = p_i(K_i - R_i) + q_i \frac{R_i}{K_i}(K_i - R_i) + r_i R_i \left(1 - \frac{R_i}{K_i}\right) - \sum_{j \neq i} \alpha_{ij} R_i R_j. \quad (4)$$

Term	Description
$p_i(K_i - R_i)$	Innovation
$q_i \frac{R_i}{K_i}(K_i - R_i)$	Imitation
$r_i R_i \left(1 - \frac{R_i}{K_i}\right)$	Natural Growth
$\sum_{j \neq i} \alpha_{ij} R_i R_j$	Competition

Table 4: Terms in the LVBC Model

Each term in the differential equation represents a different interaction, as described in Table 3.2.2. In the case of the Olympics, the **Innovation** term represents how teams improve their performance over time due to increased participation, competing in more events, and other improvements. **Imitation** may occur when a smaller team does better than expected as a result of superior coaching techniques or when achieving other seemingly unexplainable outcomes. **Natural Growth** does not occur in the Olympics context, as the number of medals available is determined by the number of events and does not change throughout the competition. Lastly, **Competition** is a measure of how well a team competes against all other teams.

Finally, we define the Adapted Lotka-Volterra-Bass Competition for the case of the Olympics. In this model, there is no natural growth of medal count, meaning $r_i = 0$ for all i . The Adapted LVBC model is defined as

$$\frac{dR_i}{dt} = p_i(K_i - R_i) + q_i \frac{R_i}{K_i}(K_i - R_i) - \sum_{j \neq i} \alpha_{ij} R_i R_j. \quad (5)$$

We define the start of an Olympics to be at $t = 0$. The countries' medal shares follow predominantly logistic growth, approaching horizontal asymptotes at either their carrying capacity or their medal count when there are no more medals left to win.

3.2.2 Scoring Model

To employ the model, we must define the values of the coefficients. We have observed that the coefficients p_i , q_i , and α_{ij} are related to the performances of the teams in the Olympics. Therefore, to figure out the coefficients, we must establish a scoring model for Olympics teams and then forecast their performance. Our first instinct is to relate economic data to the team performances;

however, we are not allowed to use any external data in this model. Therefore, we need to find alternatives. Using the provided historical data, we design a multi-dimensional scoring model for all athletes based on multiple factors. We define an athlete's personal score S as

$$S_i = w_1 \cdot H_i + w_2 \cdot A_i + w_3 \cdot E_i + w_4 \cdot G_i \quad (6)$$

Term	Description
S_i	Personal Performance Score for Athlete i
w_1, w_2, w_3, w_4	Model Weights
H_i	Historical Medal Score for Athlete i
A_i	Project Advantage Score for Athlete i 's Team
E_i	Experience Adjustment for Athlete i
G_i	Gender Compensation for Athlete i

Table 5: Terms in the Multidimensional Scoring Model

Specifically, the values of w_1, w_2, w_3, w_4 will be obtained by fitting to the historical data, and we will discuss S_i, E_i , and G_i in the later part of this section.

1. Historical Medal Score

The historical medal score quantifies a nation's past Olympic success, providing insight into its long-term performance. This is calculated using a weighted sum of medals won in previous Olympic Games:

$$H_i = \alpha \cdot \text{Golden Medal} + \beta \cdot \text{Silver Medal} + \gamma \cdot \text{Bronze Medal} \quad (7)$$

Where, $\alpha > \beta > \gamma$ reflect the relative prestige of gold, silver, and bronze medals. Historical scores are aggregated over a specific period (e.g., the past five Olympic Games) to balance recent and older performances. To prevent overemphasis on historically dominant nations, scores are normalized using Min-Max scaling. This approach highlights consistent performers while allowing emerging nations to gain competitive recognition. In our model, we assume the relative prestige of gold, silver, and bronze medals as 3, 2, and 1, respectively.

2. Project Advantage Score

Certain nations excel in specific sports due to historical dominance, technical expertise, or infrastructure investment. For example, China in table tennis or the United States in swimming. The project advantage score quantifies this domain-specific strength:

$$\text{Project Advantage Score} = \frac{\text{Medals Won by Nation in Sport}}{\text{Total Medals in Sport Across All Nations}} \quad (8)$$

This score highlights a country's relative dominance in each sport by comparing its medal count in the sport to the global total.

3. Experience Adjustment

Athletes' experience significantly impacts performance, especially in high-pressure environments like the Olympics. The model accounts for an athlete's participation in previous Olympic Games:

$$\text{Experience Score} = \alpha \cdot e - \beta \cdot e^2 + \gamma \quad (9)$$

where e represents the number of Olympic participations, and α , β , and γ are coefficients capturing the positive impact of early experience and the diminishing returns or negative effects (e.g., age-related decline) of excessive participation. This adjustment is calculated at the athlete level and aggregated for the nation's team. This approach ensures that countries with experienced athletes are credited for their expertise, while also reflecting the potential cost of over-reliance on aging participants.

4. Gender Compensation

The gender compensation adjustment acknowledges the historical underrepresentation of female athletes in many sports. Although modern Olympic Games strive for gender equality, disparities in funding, training opportunities, and representation still exist in some nations. This results in a larger range in the distribution of skill in specific female events. This adjustment ensures an accurate evaluation of a nation's overall sports system by slightly amplifying the contributions of female athletes by 5 percent. The specific value was obtained by back-testing the model against previous Olympic Games.

The results of this scoring model when predicting the results of the Paris 2024 Olympics are quite striking:

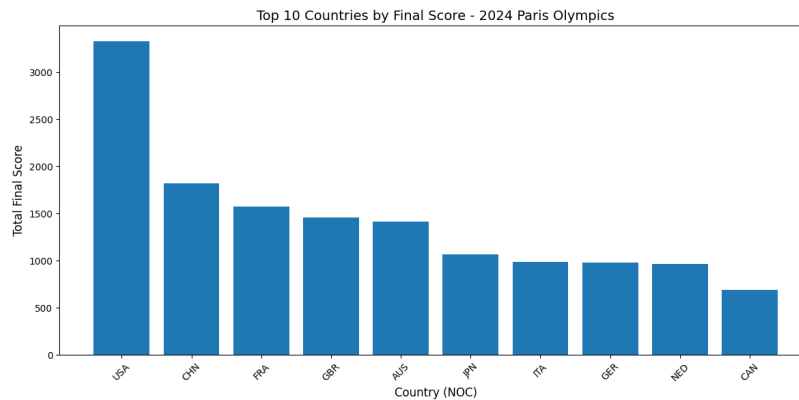


Figure 5: Countries of Top 10 Score

As shown in Figure 5, the five countries earning the highest performance scores are the USA, China, France, Great Britain, and Australia, which mostly aligns with the actual results of national medal counts in the 2024 Paris Olympics.

To extend its applicability, the model incorporates dynamic forecasting via Meta's Prophet Python package[10] to predict future scores, offering insights into potential medal distributions in upcoming games.

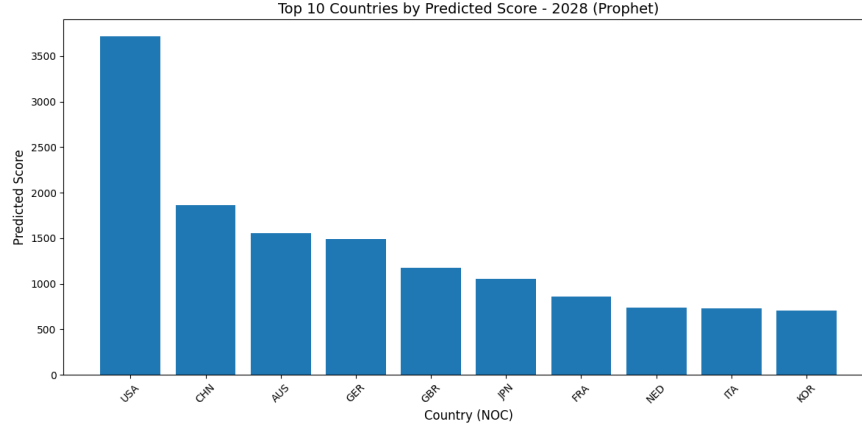


Figure 6: Prediction of Top 10 Countries

The bar chart in Figure 6 depicts the predicted scores for the top 10 countries in the Los Angeles 2028 Olympics, based on the Prophet forecasting model. The United States (USA) leads significantly with a score exceeding 3,500, followed by China (CHN) with a score above 2,000. Australia (AUS), Germany (GER), and Great Britain (GBR) occupy the next ranks, each with scores ranging from 1,000 to 1,500. This graph highlights the anticipated dominance of the USA and CHN, with a noticeable gap separating them from other contenders.

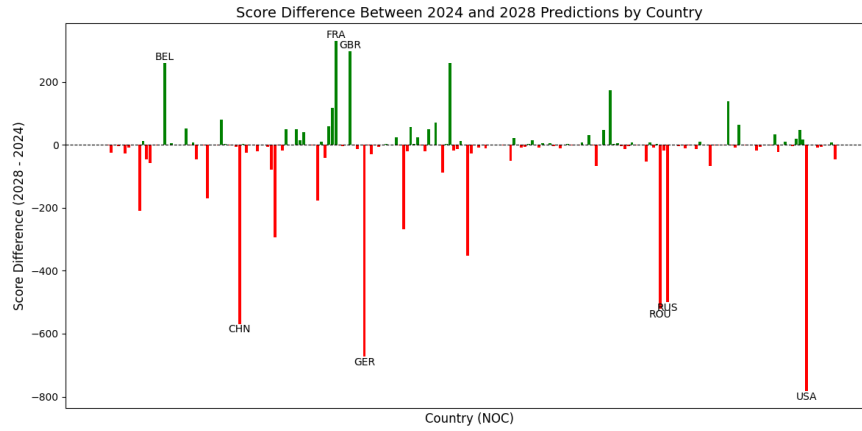


Figure 7: Residue Plot for Prediction on 2024 Paris Olympics

Finally, we define the coefficients in the Adapted LVBC model using the scoring algorithm explained above. Denoting the *standardized score* of team i as s_i , where s_i is equal to the score of team i divided by the number of athletes competing for the team, we have

- The innovation coefficient for team i is $p = (s_{24} - s_{20}) / \max(s)$, where s_{24} and s_{20} is the score of team i in 2024 and 2020, respectively, and $\max(s)$ is the highest score team i has ever achieved. The innovation coefficients take values between -1 and 1 , inclusively.
- The competition coefficient between teams i and j is $\alpha_{ij} = 1 / (|s_{28i} - s_{28j}| + 1)$. This coefficient takes values between 0 , exclusively, and 1 , inclusively.

- The (medal) carrying capacity of team i is $K = \lceil 1.4(n+1) \rceil$, where n is the amount of medals team i won in 2024. This allows teams that did not earn medals in 2024 to earn up to two medals in 2028.
- The imitation coefficient q is more difficult to define. It is hard to differentiate innovation and imitation when changing the quality of a team's performance, assuming constant competition. As will be explained in the following subsection, we define q explicitly in the model.

3.2.3 Results from the Model

Interestingly, our model is very sensitive to the choice of q , the imitation coefficient. We find that lower values ($q < 500$) lead the model to predict low medal counts, while high values ($q > 10000$) lead the model to predict every country will earn its 'carrying capacity' amount of medals. After testing many values of q , we choose a value of 2500. For small subsets of countries, this value produces the most unsurprising results. The other main assumption made is that the distribution of gold, silver, and bronze medals earned by countries in 2028 will be roughly similar to the countries' medal distributions in 2024. In other words, if a country won 5 total medals in 2024 where 4 of them are gold, we would expect the country to win mostly gold medals again in 2028.

As we are limited by computing power, we are only able to run the model on a subset of countries. Because of reasons we do not yet know, the model takes much longer to predict results for countries that have a history of winning very few medals. The countries we predict outcomes for are the top 15 medal-winners in Paris 2024. To examine the sensitivity of our model to the various coefficients and produce uncertainties in the estimations, we perform 1000 Monte-Carlo simulations, where random 'noise' is added to the q , p , and K coefficients. For the q and p coefficients, each run of the simulation adds a normally distributed random number with a mean of 0 and standard deviations of 1000 and 0.2, respectively. The values of K are scaled by normally distributed random values with a mean of 1.03 and a standard deviation of 0.1.

Rank	NOC	Gold	SD	Silver	SD	Bronze	SD	Total Medals	Total SD
1	USA	49	5	45	5	41	4	135	14
2	CHN	41	4	32	3	29	3	102	10
3	GBR	31	3	27	3	25	3	83	9
4	JPN	28	3	16	2	27	3	71	8
5	AUS	20	2	14	1	25	2	59	5
6	ITA	15	1	18	2	23	2	56	5
7	GER	17	2	13	1	19	2	49	5
8	NED	15	2	16	2	15	2	46	6
9	FRA	11	1	16	2	13	1	40	4
10	CAN	8	1	8	1	19	2	35	4

Table 6: Medal Allocation Prediction for 2024

Before predicting the 2028 results, we first run the model using data from 2020 and earlier to predict the 2024 results. This is an easy way to test if our model gives reasonable results, as well as

learn of its shortcomings in prediction. Table 6 shows the predicted medal distribution for the top 10 teams in 2024. The list is mostly correct in ranking, though does overestimate medals for some countries. The overestimation is a result of our using a subset of countries, but still every event.

Rank	NOC	Gold	SD	Silver	SD	Bronze	SD	Total Medals	Total SD
1	USA	45	5	48	5	43	4	136	14
2	CHN	46	4	35	3	26	2	107	9
3	GBR	25	2	29	3	35	3	89	8
4	FRA	19	2	27	2	24	2	70	6
5	AUS	24	2	18	2	26	3	68	7
6	JPN	28	3	16	2	18	2	62	7
7	ITA	13	1	14	1	21	2	48	4
8	NED	16	1	12	1	16	2	44	4
9	GER	12	1	13	1	13	1	38	3
10	KOR	13	1	9	1	14	1	36	3
11	CAN	11	1	9	1	14	1	34	3
12	NZL	12	1	9	1	7	1	28	3
13	BRA	6	1	8	1	11	1	25	3
14	HUN	7	1	9	1	8	1	24	3
15	ESP	5	1	8	1	10	1	23	3

Table 7: Estimated Medal Counts for 2028

After saving all the simulated values and finding how they vary, we are left with the values and uncertainties shown in Table 7 (Standard Deviation shortened to SD). In other words, the interval formed by subtracting and adding one standard deviation gives us a $\sim 68\%$ confidence interval. Interpreting the results at a 95% confidence level by creating an interval with a radius two times the standard deviation gives very reasonable prediction bounds. Interestingly, this list is similar in ranking to the 2024 Paris Olympics, but the total number of medals is a bit higher for a few teams. Specifically, our model predicts improvements in ranking for Great Britain(+4), France(+1), Italy(+2), Germany(+1), Brazil(+7). The model predicts the countries that will place worse than they did in 2024 are Japan(-3), Australia(-1), Netherlands(-2), South Korea(-2), Canada(-1), and New Zealand(-1). The model predicts the USA and China to place first and second respectively, just as they did in 2024. Hungary and Spain are also predicted to place exactly as they did in 2024, with each earning five more medals.

3.3 Advantageous Event and Medal Tally

Using the program data we generated in the scoring model, we map out the relationship between events and the corresponding most advantageous country. The graph highlights the most advantageous NOCs for various sports based on their normalized advantage scores. The Advantage Score reflects a country's historical performance, athlete experience, and contextual factors in each sport. The distribution shows that certain sports, such as archery in South Korea or basketball in the USA, are closely associated with nations that have consistently excelled due to focused investment, strong athletic traditions, or cultural prominence. The events most important for a country often align with their medal prospects, as resources are concentrated in sports where they

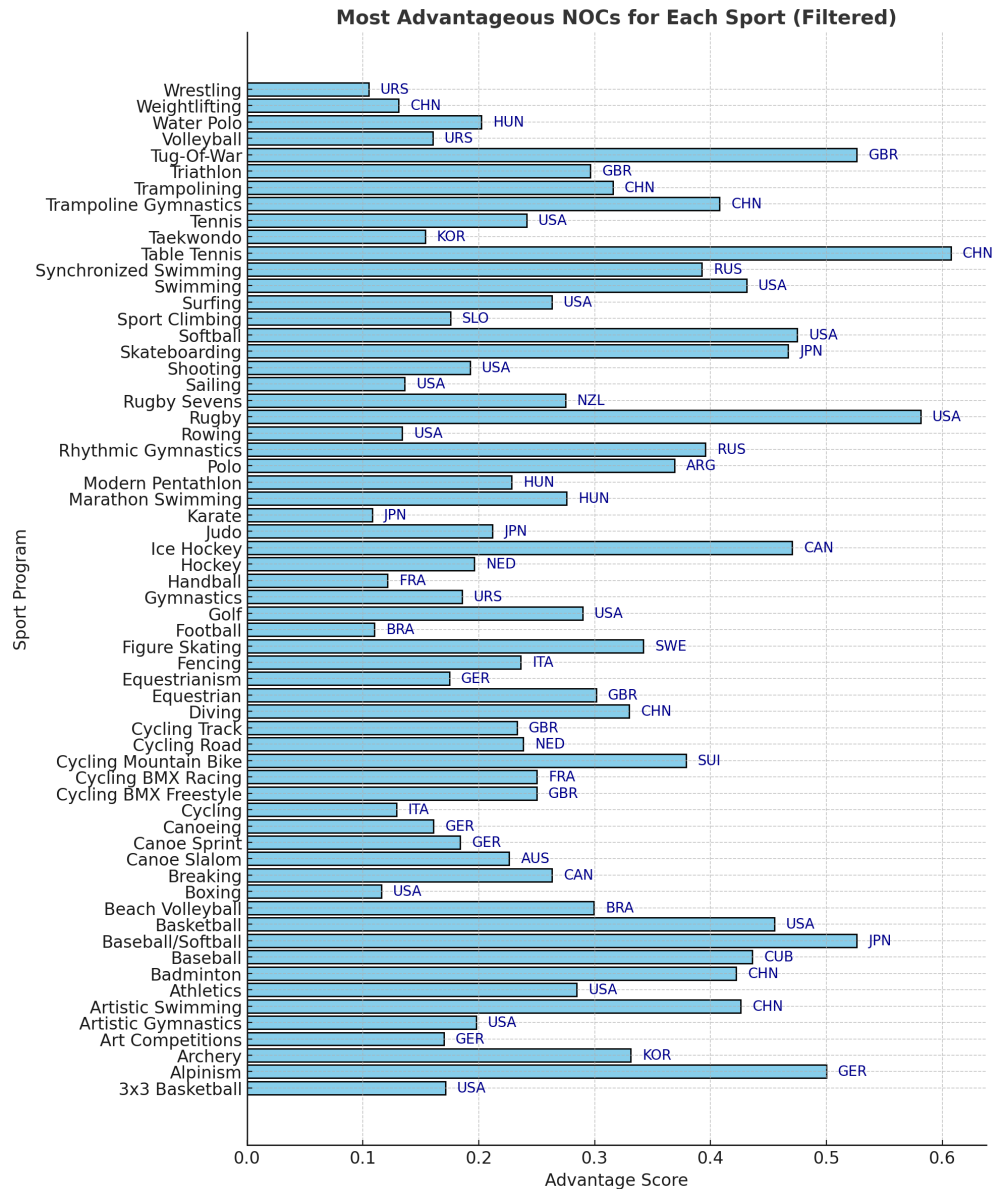


Figure 8: Event and Corresponding Most Advantageous NOC

have a competitive edge. For instance, South Korea's dominance in archery stems from a robust infrastructure and widespread participation, while the USA's basketball success reflects its global influence in the sport.

When a country hosts the Olympics, event selection can strategically impact results. Home countries often allocate resources and emphasize sports where they have historically performed well or stand a higher chance of securing medals. This focus can skew medal tallies, as host nations tend to outperform their average due to advantages like familiar conditions, local support, and reduced travel fatigue. Hosting can also boost participation and visibility in sports traditionally underrepresented in that country, fostering longer-term growth in those disciplines.

3.4 Random Forest Model

We now wish to estimate the probability that a team will win its first medal at the 2028 Olympics. To do this, we will make a crucial assumption: An increase in the number of athletes competing for a team increases the team's chance of winning their first medal. First, we consider the number of athletes that competed when a country won its first medal. We find that the median number of athletes competing when a country wins its first medal is 28. We recognize that this number is an overestimate because of the cases when large teams won medals in the 1920 Olympics, which is when our data starts, that were not necessarily the teams' first medals. As such, we define the athlete threshold as the fraction scaling the median value. To examine the sensitivity of our model, we test three different threshold values.

Our first instinct when attempting to answer this question is to employ a logistic regression model because a team may either win their first medal or not, a binary outcome. However, a logistic regression model is not appropriate in this case for a few reasons:

1. We cannot assume linearity between the number of athletes competing and the log odds of medal success. While we assume a positive relationship between the two variables, there are too many other factors to limit the analysis in such a way.
2. We cannot assume independence in medal outcomes. For example, if Team A wins gold in an event, Team B cannot also win gold in the same event. This problem would arise if most athletes on two teams are competing in the same sport.

To capture as many trends in the athlete count as possible, we use a Random Forest model on 500 trees with two variables randomly sampled at each split. The model uses features such as the year, the number of athletes competing on a team, the year-to-year growth of participants, and the rate of that growth to classify countries into two categories: those predicted to exceed the athlete threshold and those not predicted to exceed the athlete threshold. Table 8 shows the countries predicted to exceed the athlete threshold when the median is scaled by a value of 0.7. The teams, given by their NOC codes, are Angola, Guinea, and Mali.

The second threshold we test is at 0.6 times the median value. We find that the predicted first-time medalists are the same as the 0.7-scaled threshold. Finally, we test a threshold value of half the median and surprisingly, we still only predict Angola, Guinea, and Mali as the first-time medalists in 2028. By the athlete count compared to other teams, we give these three countries good odds on winning their first medals in 2028.

	NOC	Athlete Count	YoY Growth	Growth Rate	Prediction
1	ANG	26	6	1.30	Likely to win first medal
2	GUI	25	20	5.00	Likely to win first medal
3	MLI	24	20	6.00	Likely to win first medal

Table 8: Table of Predicted First-Time Medalists at a Threshold of 0.7

4 Identifying a ‘Great Coach’ Effect

4.1 Quantifying a ‘Great Coach’ Effect

Certainly, a ‘great coach’ increases the chances of a team earning medals in specific sports. In this section, we perform a case study on a famous coach Ratko Rudić. Ratko Rudić is widely regarded as one of the greatest water polo coaches in history, with a career spanning over three decades and significant tenures in multiple national teams. He led Yugoslavia to consecutive Olympic gold medals in 1984 and 1988, establishing himself as a dominant figure in the sport. His career continued with Italy, where he guided the team to Olympic gold in 1992 and bronze in 1996. Rudić’s tenure in the United States from 2001 to 2004 included a gold medal at the 2003 Pan American Games. Returning to Croatia, his homeland, he coached the national team to Olympic gold in 2012, further cementing his legacy. His final international role was with Brazil, where he led the team to the 2016 Rio Olympics. Throughout his career, Rudić’s strategic acumen and ability to transform teams made him a legendary figure in water polo.

Figure 9 demonstrates the difference between the average individual score of water polo athletes and their national team’s average score during Ratko Rudić’s coaching tenures. The horizontal black line represents the baseline where individual and national scores are equal. The green, blue, and red dashed lines represent one standard deviation (SD) above, two SDs above, and one SD below the baseline, respectively. Notably, in most years during Rudić’s tenure, the score difference exceeds two standard deviations above the baseline, signifying a significant improvement in individual athlete performance relative to the national average.

The result of a great coach is striking as well. During Rudić’s tenure (1984–1988) in Yugoslavia, the water polo team consistently performed more than two standard deviations above the baseline, resulting in consecutive Olympic gold medals. This suggests an estimated addition of 1–2 medals per Olympic cycle directly attributable to his leadership. Then from 1990–2000, Italy maintained performance levels often exceeding two SDs above the national baseline. This tenure coincided with Olympic gold in 1992 and bronze in 1996, demonstrating a similar estimated impact of 1–2 medals per cycle.

Rudić’s coaching from 2005–2012 culminated in an Olympic gold medal for Croatia in 2012.

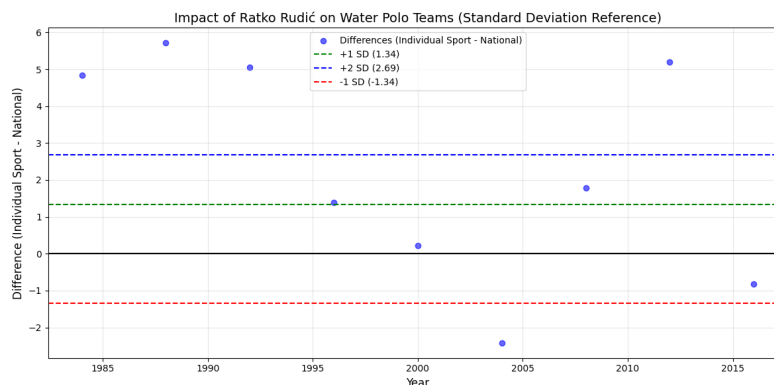


Figure 9: Impact of Ratko Rudić on Water Polo Teams

The graph shows Croatia's performance exceeding one SD above the baseline during this period, translating to an estimated impact of 1 medal per cycle. In other words, a 'great coach' might not be able to impact the medal tally of the powerful countries, but he/she is more likely bring a medal to small countries, and very likely to make a first-medalist country. However, the data also shows that coaches might not be a good fit for the team for some reasons, like Rudić's tenure with the US water polo team. Therefore, a great coach is very likely to increase the overall performance of the team with some exceptions.

4.2 Countries in Need of a 'Great Coach'

Now, we propose three countries that are in need of a 'great coach' in specific sports. First, we must define our search parameters to find countries that should, but don't, win medals in specific sports. We believe that teams with proven success and experience are more likely to be just a 'great coach effect' away from winning medals. Smaller teams with little past success might need more than just a coach to start medaling. Additionally, it is our belief that a coach has a greater impact in group, or non-individual, sports than in individual sports. This is because it is more likely a new 'star' player on a team will win a medal in an individual event than when competing in a non-individual event.

The scoring model can be effectively used to identify sports programs that would benefit from a 'great coach' by analyzing gaps in performance and potential for improvement. By calculating and comparing individual athlete scores with the national team average across various programs, the model highlights areas where individual performance consistently falls below the team average or where the national team underperforms relative to global competitors. Programs with low scores but strong foundational resources, such as athlete participation and historical performance may be prioritized. Additionally, by examining trends over multiple Olympic cycles, we can identify sports with stagnating or declining scores, suggesting a need for transformative coaching to revitalize the program. We choose three historically very competitive countries that have not won medals in specific sports in recent years and suggest programs they can invest in, as shown in Table 9. The choices of these countries and programs may also be verified with our scoring model, as we observe that for each of the teams, the indicated sports' scores are significantly below the team score.

5 Other Insights

Continuing the discussion on scoring in Section 3.2.2, we wish to find the actual reasons for our observations. To further test whether the Prophet model functions as expected, we conduct a consistency test by using data before Tokyo 2020(21) to make predictions on Paris 2024 team results and compare the predictions to the actual scores of Paris 2024 teams, the residue plot is presented in fig 7. The largest negative residuals, such as for the USA, GER (Germany), and RUS

Country	Sport Program
USA	Field Hockey
Japan	Water Polo, Rowing, Handball
Great Britain	Beach Volleyball, Artistic Swimming

Table 9: Team and Underperforming Sport(s)

(Russia), suggest these countries significantly underperformed relative to the model's predictions. Conversely, countries like BEL (Belgium), FRA (France), and GBR (Great Britain) show large positive residuals, indicating they exceeded expectations based on prior data trends.

France's notable success at the Paris 2024 Olympics can be attributed to a comprehensive strategy aimed at enhancing athletic performance[8]. The establishment of the *Agence Nationale du Sport* (ANS) in 2019 centralized expertise and directed additional funds to sports with high medal potential. This approach included hiring top-tier coaches, such as Jacco Verhaeren for swimming, and focusing on advanced training techniques. These efforts contributed to France's improved medal tally. Of course, France was also the host country in 2024.

Moreover, the UK's targeted investment approach[9], focusing on sports with high medal potential, has been instrumental in its Olympic successes. This strategy includes substantial funding for elite athletes and prioritizing resources for disciplines where British athletes have strong prospects, leading to performances that exceed model predictions.

Conversely, Germany's underperformance has been linked to underfunding and aging sports facilities[7]. Despite federal promises to increase the sports budget, the decentralized sports management and outdated infrastructure have been cited as significant factors hindering athlete development and performance.

In addition to the personnel investments, France also gained an advantage by hosting the Olympics. Familiarity with venues, the support of local crowds, and the motivation to compete on home soil can improve athlete performance. These effects contributed to France's outperformance in 2024.

6 Conclusion

In this study, we analyze historical data on every Olympic Games from 1920 to 2024 to predict the performance of teams in the 2028 Los Angeles Olympics. After preparing the data for analysis, we make a few key assumptions that help limit the complexity of working with the data. Each model used in later sections also have a few assumptions made of their own.

The most important indicators of a team's performance in the Olympics are its total and gold medal counts. As such, the key objective of the analysis is to successfully predict medal counts in the 2028 Olympics. First, we predict total and gold medal counts with a simple linear regression model. The results of this model are reasonable, even though a linear model is not an appropriate choice.

To improve on the linear regression model, we design a competition model combining aspects of the Lotka-Volterra species competition and the Bass market competition models. This model considers many more factors that contribute to a team's performance, such as a team's innovation, how competitive a team is against others, and to what degree teams 'imitate' each other. To assist in defining these parameters, we also design an algorithm that scores athletes and teams based on past performance and predicts team scores for 2028. We find very interesting results, forecasting a medal table for many countries.

We use additional approaches, such as a Random Forest model and difference equations, to address the issue of first-time medal winners and to further explore the impact of 'great coaching'.

Lastly, we perform external research to contextualize some of our observations.

Weaknesses of our approach include a simplification of bronze, silver, and gold medal dynamics. In reality, the different proportions of bronze, silver, and gold medals vary more than we account for. Another weakness is that in the Adapted LVBC model, the coefficients remain constant when in reality they should change as the Olympic Games progress.

A strength of our analysis is that we view the Olympics as a competition between species for a *fixed* amount of resources, as the total number of medals is determined at the start of the Games and does not change. Regression models fail to account for this and very easily overestimate total medal counts. Another strength of our approach is that we use a comprehensive scoring system that takes many factors into account, such as experience, event-specific dominance, and historical performance. Finally, we perform sensitivity analysis within our models that is incorporated in the uncertainty associated with every resulting value. This ensures no parameters take over the results.

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Appendices

Appendix A First appendix

This appendix contains the R code used to perform the data cleaning, various observations, and the models.

cleaningData.R

```
library(tidyverse)
data_athletes_raw = read_csv(file="./2025_Problem_C_Data/summerOly_athletes.
  ↪ csv")
data_hosts_raw = read_csv(file="./2025_Problem_C_Data/summerOly_hosts.csv")
data_medal_counts_raw = read_csv(file="./2025_Problem_C_Data/
  ↪ summerOly_medal_counts.csv")
data_programs_raw = read_csv(file="./2025_Problem_C_Data/summerOly_programs.
  ↪ csv")
## Cleaning data_programs
# Replacing NAs and non numericals with 0
#data_programs = data_programs_raw
data_programs$Sport[c(9,10)]="Baseball and Softball"
data_programs$Discipline[45]="Modern Pentathlon"
data_programs$Discipline[66]="Water Motorsports"
data_programs$Code[48]="ROQUE"
data_programs$Code[40]="JUEDEPAUME"
data_programs$'1896'[c(50,53)]=0
for (colname in colnames(data_programs)) {
# Replace weird characters with 0
data_programs[[colname]][data_programs[[colname]] == "\x95"] = 0

# Remove NAs if needed (e.g., replace with 0 or other strategy)
data_programs[[colname]][is.na(data_programs[[colname]])] = 0
}
# Remove skating and ice hockey
data_programs = data_programs |>
filter(!row_number() %in% c(70,71,72,73,74)) |>
select(-`Sports Governing Body`)
```

```

# Remove data from before 1914 (1912 Olympics and earlier)
data_programs = data_programs |>
select(-c('1896','1900','1904','1906*','1908','1912'))
# write.csv(data_programs,"clean_program_data.csv")
#####
## Cleaning athlete data
# data_athletes=data_athletes_raw
data_athletes = data_athletes |>
mutate(Medal = recode(Medal,
  "No medal" = 0,
  "Bronze" = 1,
  "Silver" = 2,
  "Gold" = 3))
data_athletes = data_athletes |>
filter(!row_number() %in% which(Year<1914))
# write.csv(data_athletes,"clean_athletes_data.csv")
#####
## Cleaning medal count data
data_medal_counts = data_medal_counts |>
mutate(NOC = str_trim(NOC))
data_medal_counts = data_medal_counts |>
filter(!row_number() %in% which(Year<1914))
# "CHL" "CEY" "BWI" "NER" "IOP" "SVN" "TON" "BRB" "KWT" "ARE" "PRY" "SDN" "TGO"
  ↳ " " "BHR" "GRD" "BWA" "GTM" "FJI" "BFA"
# replace BFA with BUR
data_medal_counts$NOC[which(data_medal_counts$NOC=="BFA")]="BUR"
# replace CHL with CHI
data_medal_counts$NOC[which(data_medal_counts$NOC=="CHL")]="CHI"
# replace CEY with SRI
data_medal_counts$NOC[which(data_medal_counts$NOC=="CEY")]="SRI"
# replace BWI with ANT
data_medal_counts$NOC[which(data_medal_counts$NOC=="BWI")]="ANT"
# replace EOR with ROT
data_medal_counts$NOC[which(data_medal_counts$NOC=="EOR")]="ROT"
# Taking too long. will just remove the 1 occurrence of the countries with old
  ↳ NOC codes in the firstMedals.R file
#write.csv(data_medal_counts,"medal_counts.csv")
#####
## Get a list of all countries that participated in 2024
countries2024 = unique(data_athletes$NOC[which(data_athletes$Year==2024)])
#write.csv(countries2024,"countries2024.csv")

```

firstMedals.R

```

haveWonMedals = unique(data_medal_counts$NOC)
allCountries= unique(data_athletes$NOC)
haveNotWonMedals = setdiff(allCountries,haveWonMedals)
# Track the year of the first competition for every team
first_competition = data_athletes |>
select(c(NOC,Year)) |>
group_by(NOC) |>
summarize(First_Olympics=min(Year))
# next want to find the years of the first medals for countries BUT not if it
  ↳ is also the first competition of the country

```

```

first_medal = data_medal_counts |>
group_by(NOC) |>
summarize(First_Medal_Year = min(Year)) # Get the first year a medal was won
first_medal$NumAthletes = NA
for (i in 1:nrow(first_medal)) {
  first_year = first_medal$First_Medal_Year[i]
  curr_NOC = first_medal$NOC[i]
  numAthletes = length(unique(data_athletes$Name[which(data_athletes$Year==
    ↪ first_year & data_athletes$NOC==curr_NOC)]))
  first_medal$NumAthletes[i] = numAthletes
}
first_medal = first_medal |>
filter(row_number() %in% which(NumAthletes>0))
medAthletesFirstMedal = round(median(first_medal$NumAthletes)*.5)
#####
# Random forest to predict athlete counts
library(randomForest)
data_non_medalist = data_athletes |>
filter(NOC %in% haveNotWonMedals)
# remove a few teams that won medals but changed their NOC
data_non_medalist = data_non_medalist |>
filter(!NOC %in% c("AIN", "EOR", "FIJ", "PAR", "SLO"))
country_athlete_counts = data_non_medalist |>
group_by(NOC, Year) |>
summarise(AthleteCount = n(), .groups = "drop")
country_athlete_counts = country_athlete_counts |>
arrange(NOC, Year) |>
group_by(NOC) |>
mutate(AthleteGrowth = AthleteCount - lag(AthleteCount, default = 0),
AthleteRate = AthleteCount / lag(AthleteCount, default = 1))
country_athlete_counts = country_athlete_counts |>
mutate(Target = ifelse(AthleteCount > medAthletesFirstMedal, 1, 0))
train_data = country_athlete_counts
rf_model = randomForest(as.factor(Target) ~ Year + AthleteCount +
  ↪ AthleteGrowth + AthleteRate,
  data = train_data,
  ntree = 500,
  mtry = 2,
  importance = TRUE)
test_data = country_athlete_counts |>
filter(Year == 2024 & NOC %in% haveNotWonMedals) |>
mutate(Year = 2028)
test_data$Predicted = predict(rf_model, newdata = test_data, type = "response
  ↪ ")
test_data = test_data |>
mutate(PredictedWin = ifelse(Predicted == 1, "Likely to win first medal", "
  ↪ Unlikely to win first medal"))
medalTable = test_data |>
filter(Predicted==1)|>
select(NOC,AthleteCount,AthleteGrowth,AthleteRate,PredictedWin)

```

LVBC.R

total_gold = 339

```

total_silver = 338
total_bronze = 395
q_value = 2500
selected_NOCs = c("USA", "CHN", "GBR", "GER", "FRA", "JPN", "AUS",
"NED", "KOR", "ITA", "NZL", "CAN", "BRA", "HUN", "ESP")
innovationP_subset = innovationP |> filter(NOC %in% selected_NOCs)
competitionMatrix_subset = competitionMatrix[selected_NOCs, selected_NOCs]
carrying_capacity_subset = carrying_capacity |> filter(NOC %in% selected_NOCs)
NOCs = innovationP_subset$NOC
p_values = setNames(innovationP_subset$p, innovationP_subset$NOC)
K_values = setNames(carrying_capacity_subset$K, carrying_capacity_subset$NOC)
if (any(K_values <= 0)) stop("Error: Carrying capacities must be greater than
  ↳ zero.")
initial_R = setNames(rep(1, length(NOCs)), NOCs)
alvbc_model = function(t, R, params) {
p = params$p
K = params$K
alpha = params$alpha
q = params$q
dRdt = numeric(length(R))
for (i in seq_along(R)) {
innovation = p[i] * (K[i] - R[i])
imitation = q * (R[i] / K[i]) * (K[i] - R[i])
competition = sum(alpha[i, -i] * R[i] * R[-i])
dRdt[i] = innovation + imitation - competition
}return(list(dRdt))}
params = list(p = p_values, K = K_values, alpha = competitionMatrix_subset, q
  ↳ = q_value)
n_ticks = 100
time = seq(0, n_ticks, by = 1)
ode_solution = ode(y = initial_R, times = time, func = alvbc_model, parms =
  ↳ params, method = "lsoda")
final_row = ode_solution[nrow(ode_solution), -1]
predicted_medals = data.frame(NOC = NOCs, Medal_Count = as.numeric(final_row))
predicted_medals = predicted_medals |> arrange(desc(Medal_Count))
recent_medals = medal_counts |> filter(Year %in% c(2020, 2024))
medal_proportions = recent_medals |>
group_by(NOC) |>
summarise(
Total = sum(Gold + Silver + Bronze),
Gold_Prop = sum(Gold) / Total,
Silver_Prop = sum(Silver) / Total,
Bronze_Prop = sum(Bronze) / Total
) |>
replace_na(list(Gold_Prop = 0, Silver_Prop = 0, Bronze_Prop = 0))
predicted_medals = predicted_medals |>
left_join(medal_proportions, by = "NOC") |>
mutate(
Gold = round(Medal_Count * Gold_Prop),
Silver = round(Medal_Count * Silver_Prop),
Bronze = round(Medal_Count * Bronze_Prop))
predicted_medals$Gold = round(predicted_medals$Gold * total_gold / sum(
  ↳ predicted_medals$Gold))
predicted_medals$Silver = round(predicted_medals$Silver * total_silver / sum(

```



```

    ↪ predicted_medals$Silver))
predicted_medals$Bronze = round(predicted_medals$Bronze * total_bronze / sum(
    ↪ predicted_medals$Bronze))
gold_results_matrix = matrix(NA, nrow = length(NOCs), ncol = num_simulations)
silver_results_matrix = matrix(NA, nrow = length(NOCs), ncol = num_simulations
    ↪ )
bronze_results_matrix = matrix(NA, nrow = length(NOCs), ncol = num_simulations
    ↪ )
num_simulations = 1000
predicted_medals = predicted_medals |> arrange(NOC)
medal_proportions = medal_proportions |> arrange(NOC)
gold_results_matrix = matrix(NA, nrow = length(NOCs), ncol = num_simulations)
silver_results_matrix = matrix(NA, nrow = length(NOCs), ncol = num_simulations
    ↪ )
bronze_results_matrix = matrix(NA, nrow = length(NOCs), ncol = num_simulations
    ↪ )
for (i in 1:num_simulations) {
  noisy_q = q_value + rnorm(1, mean = 0, sd = 1000) # Noise for q
  noisy_p_values = p_values + rnorm(length(p_values), mean = 0, sd = 0.2) #
    ↪ Noise for p values
  noisy_K_values = round(K_values*(rnorm(length(K_values), mean=1.03, sd=0.1)))
  params_noisy = list(p = noisy_p_values, K = noisy_K_values, alpha =
    ↪ competitionMatrix_subset, q = noisy_q)
  ode_solution_noisy = ode(y = initial_R, times = time, func = alvbc_model,
    ↪ parms = params_noisy, method = "lsoda")
  final_row_noisy = ode_solution_noisy[nrow(ode_solution_noisy), -1]
  predicted_medals_noisy = data.frame(NOC = NOCs, Medal_Count = as.numeric(
    ↪ final_row_noisy))
  predicted_medals_noisy = predicted_medals_noisy |> arrange(NOC) # Ensure
    ↪ order matches NOCs
  predicted_medals_noisy = predicted_medals_noisy |>
  left_join(medal_proportions, by = "NOC") |>
  mutate(
    Gold_Allocated = round(Medal_Count * Gold_Prop),
    Silver_Allocated = round(Medal_Count * Silver_Prop),
    Bronze_Allocated = round(Medal_Count * Bronze_Prop))
  gold_results_matrix[, i] = predicted_medals_noisy$Gold_Allocated
  silver_results_matrix[, i] = predicted_medals_noisy$Silver_Allocated
  bronze_results_matrix[, i] = predicted_medals_noisy$Bronze_Allocated}
gold_sd = apply(gold_results_matrix, 1, sd, na.rm = TRUE)
silver_sd = apply(silver_results_matrix, 1, sd, na.rm = TRUE)
bronze_sd = apply(bronze_results_matrix, 1, sd, na.rm = TRUE)
final_results = data.frame(
  NOC = NOCs,
  Gold = rowMeans(gold_results_matrix, na.rm = TRUE),
  Gold_SD = gold_sd,
  Silver = rowMeans(silver_results_matrix, na.rm = TRUE),
  Silver_SD = silver_sd,
  Bronze = rowMeans(bronze_results_matrix, na.rm = TRUE),
  Bronze_SD = bronze_sd)

```

Report on Use of AI

1. OpenAI (GPT-4)

Query1: Write r code with dplyr that will replace each of the repeated names in a column data_medal_counts\$NOC with the version without the trailing space

Output: You can use the mutate() function in combination with str_trim() from the stringr package to remove trailing spaces from the NOC column in your data_medal_counts tibble. Here's the code using dplyr and stringr:

```
library(dplyr)
library(stringr)

data_medal_counts <- data_medal_counts %>%
  mutate(NOC = str_trim(NOC))
```

2. OpenAI (GPT-4)

Query1: What predator prey model can be used to model olympics medal predictions for each country? the model should have some kind of constants that can be determined based on historical data, such as how well one country competes against another or how well a country typically does

Output: To model Olympic medal predictions for each country using a predator-prey model, we can adapt the classic Lotka-Volterra predator-prey equations to represent competition for medals. In this context, countries can be seen as "predators" (competing for medals) and "prey" (representing the events or competitions), where the success or failure of one country in winning medals affects the medal prospects of the other countries. The constants in the model can be determined from historical data, such as historical medal counts, competition strength, and how well each country competes in different events or against each other.

3. OpenAI (GPT-4)

Query1: given the r code i now want to figure out which of the countries in the created list have not won a medal in a sport in at least 7 past olympics (2024,2020....). i want to know the list of countries and also the list of sports. use the data file data_athletes which has columns Name, NOC, Year, Sport, and Medal (0 for no medal, 1 for bronze, 2 for silver, 3 for gold)

```
validCountries = data_medal_counts |>
  filter(Year==2024) |>
  select(NOC)

top_10_countries = data_medal_counts |>
  filter(Rank <= 10) |>
  filter(Year>1950) |>
  filter(NOC %in% validCountries$NOC) |>
  group_by(NOC) |>
  summarise(top_10_count = n_distinct(Year)) |>
  filter(top_10_count >= 5)

countries_in_top_10_5_or_more = top_10_countries$NOC
```

```
# Now want to consider teams that have not won a medal in any single sport
  ↳ in the past 7 Olympics
```

Output: Here is the R code to find the list of countries that have not won a medal in a sport in at least the past 7 Olympics (2024, 2020, ...), and to also identify the list of sports where they have not won a medal.

```
library(dplyr)

# Define the last 7 Olympic years
recent_olympics <- c(2024, 2020, 2016, 2012, 2008, 2004, 2000)

# Filter the athlete data for relevant countries and recent Olympics
recent_medals <- data_athletes %>%
  filter(NOC %in% countries_in_top_10_5_or_more & Year %in% recent_olympics
  ↳ )

# Identify countries and sports where medals were won
medal_wins <- recent_medals %>%
  filter(Medal > 0) %>% # Consider only cases where medals were won
  group_by(NOC, Sport) %>%
  summarise(HasMedal = n(), .groups = "drop")

# Find countries and sports with no medals in the past 7 Olympics
countries_sports_no_medals <- recent_medals %>%
  anti_join(medal_wins, by = c("NOC", "Sport")) %>% # Exclude medal-
  ↳ winning records
  distinct(NOC, Sport)

# Get the list of countries that have not won a medal in any sport for 7
  ↳ Olympics
countries_no_medals_list <- countries_sports_no_medals %>%
  group_by(NOC) %>%
  summarise(sports_without_medals = n_distinct(Sport), .groups = "drop")
  ↳ %>%
  filter(sports_without_medals > 0) %>%
  pull(NOC)

# Get the list of sports where these countries have not won medals
sports_no_medals_list <- countries_sports_no_medals %>%
  filter(NOC %in% countries_no_medals_list) %>%
  distinct(Sport) %>%
  pull(Sport)

# Print results
print("Countries without medals in any sport for past 7 Olympics:")
print(countries_no_medals_list)

print("List of sports where these countries have not won medals:")
print(sports_no_medals_list)
```

Query2: give r code defining a vector teamSports that are all the sports from the following list that typically involve at least 3 players: [1] "Diving" "Wrestling" "Rowing" "Basketball"

[5] "Triathlon" "Water Polo" "Judo" "Fencing" [9] "Synchronized Swimming" "Trampolining" "Softball" "Shooting" [13] "Hockey" "Football" "Volleyball" "Badminton" [17] "Beach Volleyball" "Handball" "Athletics" "Table Tennis" [21] "Archery" "Equestrianism" "Tennis" "Baseball" [25] "Rhythmic Gymnastics" "Weightlifting" "Cycling" "Swimming" [29] "Golf" "Gymnastics" "Canoeing" "Modern Pentathlon" [33] "Rugby Sevens" "Sailing" "Taekwondo" "Canoe Slalom" [37] "Trampoline Gymnastics" "Artistic Gymnastics" "Marathon Swimming" "Cycling Road" [41] "Equestrian" "Canoe Sprint" "Karate" "Cycling Mountain Bike" [45] "Skateboarding" "3x3 Basketball" "Artistic Swimming" "Baseball/Softball" [49] "Sport Climbing" "Cycling BMX Racing" "Surfing" "Cycling Track" [53] "Cycling BMX Freestyle" "Breaking" "Cycling Road, Cycling Track" "Cycling Road, Cycling Mountain Bike" [57] "Marathon Swimming, Swimming" "Cycling Road, Triathlon" "3x3 Basketball, Basketball" Output: Here's the R code to define a vector teamSports that contains all the sports from the given list that typically involve at least 3 players:

```
# Define the full list of sports
sports_list <- c("Diving", "Wrestling", "Rowing", "Basketball", "Triathlon"
  ↪ , "Water Polo",
    "Judo", "Fencing", "Synchronized Swimming", "Trampolining"
  ↪ , "Softball", "Shooting",
    "Hockey", "Football", "Volleyball", "Badminton", "Beach
  ↪ Volleyball", "Handball",
    "Athletics", "Table Tennis", "Archery", "Equestrianism", "
  ↪ Tennis", "Baseball",
    "Rhythmic Gymnastics", "Weightlifting", "Cycling", "
  ↪ Swimming", "Golf", "Gymnastics",
    "Canoeing", "Modern Pentathlon", "Rugby Sevens", "Sailing"
  ↪ , "Taekwondo", "Canoe Slalom",
    "Trampoline Gymnastics", "Artistic Gymnastics", "Marathon
  ↪ Swimming", "Cycling Road",
    "Equestrian", "Canoe Sprint", "Karate", "Cycling Mountain
  ↪ Bike", "Skateboarding",
    "3x3 Basketball", "Artistic Swimming", "Baseball/Softball"
  ↪ , "Sport Climbing",
    "Cycling BMX Racing", "Surfing", "Cycling Track", "Cycling
  ↪ BMX Freestyle", "Breaking",
    "Cycling Road, Cycling Track", "Cycling Road, Cycling
  ↪ Mountain Bike",
    "Marathon Swimming, Swimming", "Cycling Road, Triathlon",
  ↪ "3x3 Basketball, Basketball")

# Filter sports that typically involve at least 3 players
teamSports <- c("Basketball", "Water Polo", "Synchronized Swimming", "
  ↪ Softball", "Hockey",
    "Football", "Volleyball", "Beach Volleyball", "Handball", "
  ↪ Rowing", "Rugby Sevens",
    "Artistic Swimming", "Baseball", "Baseball/Softball", "3x3
  ↪ Basketball", "3x3 Basketball, Basketball")

# Print the selected team sports
print(teamSports)
```

OpenAI (GPT-4o) Query1: To further test whether the Prophet model functions as expected, we conduct a consistency test by using data before Tokyo 2020(21) to make prediction on Paris 2024 team results and compare the predictions to the actual scores of Paris 2024 teams, the residue plot is presented in fig 7. The largest negative residuals, such as for the USA, GER (Germany), and RUS (Russia), suggest these countries significantly underperformed relative to the model's predictions. Conversely, countries like BEL (Belgium), FRA (France), and GBR (Great Britain) show large positive residuals, indicating they exceeded expectations based on prior data trends. Any news can be used to interpret these residues?

Output: The residuals observed in your analysis, where countries like the USA, Germany, and Russia underperformed, while Belgium, France, and Great Britain exceeded expectations at the Paris 2024 Olympics, can be interpreted through various factors:

Underperformance:

United States: Despite leading the overall medal count, the U.S. men's swimming team secured only one individual gold medal, won by Bobby Finke in the 1500-meter freestyle. This was considered an underperformance compared to previous Olympics. PEOPLE.COM

Germany: Germany experienced its worst Olympic performance since reunification, finishing 10th with 33 medals. Contributing factors included underfunding, decentralized sports management, and aging facilities. LE MONDE.FR

Russia: Russia's participation was significantly impacted by sanctions and exclusions due to geopolitical issues, leading to a smaller team and reduced medal prospects. NEW AMERICA

Overperformance:

Belgium: Belgium achieved notable successes, including Remco Evenepoel's historic wins in both the time trial and road race cycling events. THE TIMES & THE SUNDAY TIMES

France: As the host nation, France leveraged home advantage to secure 64 medals, nearly doubling their Tokyo 2020 total. Strategic investments and centralized efforts contributed to this success. FT.COM

Great Britain: Team GB amassed 65 medals, with standout performances in rowing, cycling, and triathlon. Despite winning 14 golds—their lowest in 20 years—the total medal count was substantial. TALKSPORT

These outcomes highlight the complex interplay of factors such as funding, infrastructure, geopolitical issues, and strategic planning in determining Olympic performance.